

The entropy photon-number inequality

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Abstract

We prove the entropy photon-number inequality for two independent bosonic inputs with finite mean energy, for every finite number of modes. This resolves the entropy photon-number conjecture in this finite-energy setting while allowing arbitrary entanglement among the modes within either input. As a consequence, we determine the exact minimum output entropy at fixed input entropy for every finite tensor power of an identical thermal attenuator, over finite-energy inputs that may be entangled across modes. We also obtain the exact classical capacity region of the degraded two-receiver pure-loss bosonic broadcast channel under a mean photon constraint.

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1 Introduction

The entropy photon number of an n -mode bosonic state is the mean photon number per mode of the product of n identical one-mode thermal states with the same total entropy. Guha, Erkmen, and Shapiro conjectured that this quantity obeys a concavity inequality under beam-splitter mixing of independent inputs [20]. Their entropy photon-number inequality is a quantum counterpart of the classical entropy power inequality and has consequences for minimum output entropy and the information capacities of bosonic channels. We prove its two-input form for finite-energy states.

Write

$$g(t) = (t+1) \log(t+1) - t \log t, \quad t \geq 0, \quad (1.1)$$

with $0 \log 0 = 0$. This continuous function increases from zero to infinity; its inverse is denoted by g^{-1} . All entropies and logarithms in this paper use the natural base.

Theorem 1.1 (Entropy photon-number inequality). *Let $n \geq 1$ be finite, and let ρ_A, ρ_B be states on n bosonic modes, with annihilation operators $a_1, \dots, a_n$ and $b_1, \dots, b_n$, respectively, satisfying*

$$\mathrm{Tr} \rho_A \sum_{j=1}^n a_j^\dagger a_j < \infty, \quad \mathrm{Tr} \rho_B \sum_{j=1}^n b_j^\dagger b_j < \infty.$$

The joint input is $\rho_A \otimes \rho_B$; no independence assumption is made among the modes within either state. For $0 \leq \eta \leq 1$, let ρ_C be the reduced state of the modes

$$c_j = \sqrt{\eta} a_j + \sqrt{1-\eta} b_j, \quad 1 \leq j \leq n,$$

after passive beam-splitter mixing. Then

$$g^{-1}\left(\frac{S(\rho_C)}{n}\right) \geq \eta g^{-1}\left(\frac{S(\rho_A)}{n}\right) + (1-\eta) g^{-1}\left(\frac{S(\rho_B)}{n}\right), \quad (1.2)$$

where $S(\rho) = -\mathrm{Tr} \rho \log \rho$.

Thus Theorem 1.1 resolves the entropy photon-number conjecture positively for finite-energy inputs, including arbitrary internal multimode entanglement. Thermal inputs with a common mean photon number within each port give equality, even when the means at the two ports differ.

For the channel consequences, write τ_r for the n -mode product thermal state with mean photon number r in each mode, including the vacuum at $r = 0$; thus $S(\tau_r) = ng(r)$. Let $\mathcal{E}_{\eta, N_B}$ be the one-mode thermal attenuator obtained by retaining $c = \sqrt{\eta} a + \sqrt{1-\eta} b$ when the environment mode b is thermal with mean photon number N_B and is independent of the input. Its tensor power $\mathcal{E}_{\eta, N_B}^{\otimes n}$ uses the same parameters in every mode and the joint input $\rho \otimes \tau_{N_B}$.

Corollary 1.2 (Constrained entropy minimum for identical thermal attenuators). *Let $n \geq 1$ be finite, let $s, N_B \geq 0$ be finite, and let $0 \leq \eta \leq 1$. Every state ρ on n modes with $\mathrm{Tr} \rho \sum_{j=1}^n a_j^\dagger a_j < \infty$ satisfies*

$$\frac{1}{n} S\left(\mathcal{E}_{\eta, N_B}^{\otimes n}(\rho)\right) \geq g\left(\eta g^{-1}\left(\frac{S(\rho)}{n}\right) + (1-\eta) N_B\right). \quad (1.3)$$

In particular, the exact constrained minimum is

$$\min_{\substack{\rho \text{ a state on } n \text{ modes } n \\ \mathrm{Tr} \rho \sum_{j=1}^n a_j^\dagger a_j < \infty \\ S(\rho) = ns}} \frac{1}{n} S\left(\mathcal{E}_{\eta, N_B}^{\otimes n}(\rho)\right) = g\left(\eta g^{-1}(s) + (1-\eta) N_B\right). \quad (1.4)$$

It is attained by the product thermal input $\tau_{g^{-1}(s)}$. No independence assumption is imposed among the modes of ρ ; only the environment is required to be independent of that input.

Proof. The environment τ_{N_B} has finite total mean photon number nN_B and entropy $ng(N_B)$. The retained output has finite energy because passive mixing preserves combined total number, so its entropy is finite by Lemma 2.2. Apply Theorem 1.1 to the independent inputs ρ and τ_{N_B} , and then apply the increasing function g , to obtain (1.3).

For $r = g^{-1}(s) < \infty$, the state τ_r has finite total mean photon number nr and entropy ns . Proposition 2.3 gives

$$\mathcal{E}_{\eta, N_B}^{\otimes n}(\tau_r) = \tau_{\eta r + (1-\eta)N_B},$$

whose entropy is $ng(\eta r + (1-\eta)N_B)$. This attains the lower bound and proves (1.4). The endpoints are included: at $\eta = 0$ the retained state is the environment, and at $\eta = 1$ it is the input. When $s = 0$ or $N_B = 0$, the corresponding thermal state is the vacuum, covered by the zero-mean case of Proposition 2.3. $\square$

Corollary 1.2 concerns identical passive thermal attenuators with an independent product thermal environment. It asserts attainment, not uniqueness, and makes no claim for mode-dependent attenuations, amplifiers, or additive-noise channels.

The vacuum case also determines the classical capacity region of a degraded pure-loss broadcast channel. Here one input mode per use is passively split with auxiliary vacuum modes independent of the input between receivers B, C and an inaccessible loss output. The power transmissivities satisfy $0 \leq \eta_C < \eta_B$ and $\eta_B + \eta_C \leq 1$, so C 's marginal is obtained from B 's by a further pure-loss attenuation. Section 7 gives the exact region for independent classical messages and separate collective decoding, without feedback or receiver cooperation. Its finite mean photon constraint is averaged over the codebook and uses, and its finite-energy codewords may be entangled across uses. The vacuum-port specialization of Theorem 1.1 supplies the entropy input assumed in the converse of Guha, Shapiro, and Erkmén [21]; coherent-state superposition coding supplies achievability.

1.1 Context and earlier results

Shannon introduced entropy power in his theory of communication [28]; the Fisher-information proofs of Stam and Blachman [30, 4] became central to its later development. For independent classical random vectors, the entropy power inequality compares the entropy of a sum with the entropies of its summands. Its bosonic counterpart replaces addition by passive mixing of quantum modes. The analogy is especially natural for thermal states: their mean photon numbers add with the beam-splitter weights, and their entropies are given by the same function g appearing in Theorem 1.1. The formulation of Guha, Erkmén, and Shapiro [20, Section II.B] makes this thermal comparison the central conjecture and explains its consequences for bosonic communication. Earlier work had established the capacity of the pure-loss bosonic channel [18] and connected noisy-channel capacity questions with minimum output entropy [17]. The constrained minimum-output problem also appears in the bosonic broadcast-channel analysis of Guha, Shapiro, and Erkmén [21]. Early EPnI special cases included one-mode number-diagonal inputs with two nonzero probabilities and a vacuum second port [6].

König and Smith [24] established a quantum entropy power inequality using quantum Fisher information and a heat evolution. Their results include the linear bound

$$S(\rho_C) \geq \eta S(\rho_A) + (1-\eta)S(\rho_B)$$

for every transmissivity, and the exponential entropy power bound for a balanced beam splitter. De Palma, Mari, and Giovannetti [10] proved the exponential bound for every transmissivity,

$$e^{S(\rho_C)/n} \geq \eta e^{S(\rho_A)/n} + (1-\eta)e^{S(\rho_B)/n}. \quad (1.5)$$

These results are formulated for an arbitrary finite number of modes and independent inputs, with no product assumption within a group. De Palma, Mari, Lloyd, and Giovannetti [11] subsequently treated more general linear mixing of several independent groups of modes. De Palma and Trevisan later gave a rigorous finite-energy treatment of the Fisher-information and heat-flow steps using an integral formulation; their conditional inequality with a trivial memory system yields (1.5) for arbitrary independent finite-energy inputs [12, Theorem 6].

The entropy photon-number inequality requires a sharper comparison than (1.5) when the input entropy levels differ. When $S(\rho_A) = S(\rho_B)$, the linear entropy bound already gives the EPnI value by monotonicity of g^{-1} . For unequal entropy levels, thermal inputs attain EPnI, whereas the exponential bound does not recover their exact output entropy. This distinction matters for the minimum output entropy problem with an input-entropy constraint: the bound must reproduce the exact thermal comparison at unequal entropy levels.

The EPnI itself was also known for two independent Gaussian inputs in any finite number of modes, including Gaussian correlations within either input; De Palma proves this case using symplectic eigenvalue comparisons [7, Appendix A.8].

The Gaussian optimizer theorem of Giovannetti, Holevo, and García-Patrón [19] settled the unconstrained minimum-output problem for broad classes of Gaussian channels. Fixing a positive input entropy is a different optimization problem. For one-mode gauge-covariant channels, De Palma, Trevisan, and Giovannetti established optimality of passive rearrangement at fixed spectrum [13], then the sharp entropy bound for the quantum-limited attenuator [14], and finally the sharp bound for one-mode phase-covariant Gaussian channels with thermal noise [15, Theorem 4]. Their multimode extension [16, Theorem 11] covers inputs diagonal in some product basis, including classical correlations in that basis.

Later advances address further parts of this landscape. De Palma [8] proved the sharp constrained bound for multimode entanglement-breaking Gaussian attenuators and amplifiers with thermal environments, and improved lower bounds for other parameter regimes. Beigi and Rahimi-Keshari [3] proved a one-mode meta-logarithmic-Sobolev inequality that also gives a variational proof of the one-mode constrained minimum-output result. Beigi and Mehrabi [2] established subset-convolution inequalities for exponential entropy power and entropy monotonicity under repeated symmetric quantum convolution of an identical input.

Theorem 1.1 supplies the thermal comparison for two freely varying inputs, including arbitrary entanglement within either port. Corollary 1.2 specializes it to the exact finite-mode constrained minimum for identical thermal attenuators, without a product or separability restriction on the finite-energy input. Thermal states enter the proof of Theorem 1.1 as reference states and as limits of regularized minimizers. The original inputs need not be Gaussian, number-diagonal, or independent across modes.

1.2 Proof strategy

The difficulty is to compare the two freely varying inputs with thermal states at their respective entropy levels. We fix those thermal reference parameters near a hypothetical strictly violating pair. This turns the contradiction into a negative minimum of a linear combination of three entropies, after adding penalties that ensure compactness and regularity. The reference parameters remain fixed throughout the variational argument.

An independent analytic ingredient is an interpolation theorem for positive diagonal quadratic forms, proved in Section 3. For positive scalars m and real parameters x, y , it transforms four

comparisons with weights

$$mf(x)e^x, \quad mf(x)e^{-x}, \quad mf(y)e^y, \quad mf(y)e^{-y}, \quad f(x) = \frac{x}{\sinh x}, \quad f(0) = 1,$$

into the comparison with weight $mf(x)f(y)/f(x+y)$. No compatibility is required between the linear maps and the diagonal multipliers. A Stieltjes representation for the derivative of an implicitly defined function is the essential analytic step. The theorem is formulated for finitely many spectral parameter values on source tests and permits unrestricted nonnegative target sums, so it also applies to the infinite matrix spaces used below.

Section 4 develops the variational argument. Assuming a strict violation, we add a thermal port, a small thermal replacement of the output, a relative-entropy penalty, and a fourth-moment penalty. The resulting functional has faithful Gibbs minimizers with enough moment control for all subsequent traces. Varying one input at a time yields a component contraction in the logarithmic-mean metric *at those minimizers*. It is not asserted as a universal channel inequality.

Section 5 combines two product-port variance estimates with the component contractions. These supply precisely the four hypotheses of the interpolation theorem. Its conclusion controls the norm of a centered thermal-balance defect: a linear functional measuring the creation–annihilation balance after subtracting the state’s mean. In Section 6, pairing the Gibbs identities with the relaxation generators gives an exact cancellation of the centered energy terms. As the fourth-moment penalty vanishes, the remaining defects and means tend to zero; their number-basis identities force both limiting inputs to be thermal. Entropy continuity then contradicts the original strictly negative objective value.

The auxiliary regularizations have linked roles: the relative-entropy penalty gives energy coercivity and supplies the strict margin in the final defect and mean estimate, while the thermal port supplies the slack needed for the associated Hessian comparisons. Output replacement controls the output logarithm, and the moment penalty justifies the unbounded-generator calculations. Only the last penalty tends to zero in the concluding compactness argument. This avoids assuming differentiability of entropy along an unbounded evolution or convergence of the input energies.

Section 2 supplies the entropy, compactness, and thermal-mixing facts used in this argument. Each of the specialized interpolation, minimization, and limiting statements is proved within the paper.

2 Bosonic states, energy, and entropy

We work on the n -mode Fock space $\mathcal{H}_n = \ell^2(\mathbb{N}_0^n)$, with number basis $\{|\mathbf{k}\rangle : \mathbf{k} \in \mathbb{N}_0^n\}$. For the annihilation operators d_j ,

$$d_j|\mathbf{k}\rangle = \sqrt{k_j}|\mathbf{k} - \mathbf{e}_j\rangle, \quad N = \sum_{j=1}^n d_j^\dagger d_j, \quad W = I + N.$$

A state is a positive trace-class operator of trace one. Expectations of positive unbounded operators are understood as increasing limits of bounded spectral truncations. Form inequalities have their usual quadratic-form meaning on the indicated domains.

For $r > 0$, put

$$h(r) = \log(1 + 1/r), \quad \tau_r = (r + 1)^{-n} e^{-h(r)N}. \quad (2.1)$$

The product geometric law gives

$$\mathrm{Tr} \tau_r N = nr, \quad S(\tau_r) = ng(r). \quad (2.2)$$

The state τ_r is faithful and has all number moments finite. At $r = 0$, its limiting state is the multimode vacuum.

2.1 Entropy estimates and compactness

We first record the entropy facts needed in the infinite-dimensional argument. The relative entropy introduced by Umegaki [31] gives the Gibbs variational principle. For oscillator energy bounds, the compactness and entropy-continuity statements below are special cases of the general energy-constrained theory [29, 32]; we include the elementary cutoff proof used here.

Lemma 2.1 (Relative entropy and the Gibbs variational inequality). *Let σ be a faithful state, and suppose both $S(\rho)$ and $\text{Tr } \rho(-\log \sigma)$ are finite. Then*

$$D(\rho \| \sigma) := -S(\rho) + \text{Tr } \rho(-\log \sigma) \geq 0,$$

with equality if and only if $\rho = \sigma$. Consequently, if K is bounded below and $0 < Z = \text{Tr } e^{-K} < \infty$, its Gibbs state $Z^{-1}e^{-K}$ uniquely minimizes $\text{Tr } \rho K - S(\rho)$ among states for which these expressions are finite, provided that Gibbs state belongs to the admitted class.

Proof. Choose eigenvectors ψ_j of ρ with positive eigenvalues p_j , and set $q_j = \langle \psi_j, \sigma \psi_j \rangle$. Scalar Jensen applied to the spectral measure of σ gives

$$-\langle \psi_j, \log \sigma \psi_j \rangle \geq -\log q_j.$$

The elementary inequality $p \log(p/q) \geq p - q$ and $\sum_j q_j \leq 1$ now imply nonnegativity. Equality requires $p_j = q_j$, no mass of σ outside the support of ρ , and equality in each Jensen inequality. The last condition makes each ψ_j an eigenvector of σ with eigenvalue q_j , so $\rho = \sigma$. The Gibbs assertion follows by substituting $-\log \sigma = K + \log Z$. $\square$

Lemma 2.2 (Energy bound and compactness). *For each finite $E \geq 0$, the set of states with $\text{Tr } \rho N \leq E$ is compact in trace norm. On this set entropy is continuous and obeys*

$$0 \leq S(\rho) \leq ng(E/n). \quad (2.3)$$

Every finite-energy state is a trace-norm limit of normalized finite-number truncations with uniformly bounded energies and convergent entropies.

Proof. Diagonal entropy in any orthonormal basis bounds von Neumann entropy above: apply concavity of $-t \log t$ to each diagonal entry expressed in a spectral decomposition, then sum the nonnegative terms. Compare the number-basis probabilities with the geometric probabilities of $\tau_{E/n}$. Summing $-p \log p \leq -p \log q + q - p$ gives, when $E > 0$,

$$S(\rho) \leq n \log(1 + E/n) + h(E/n)E = ng(E/n).$$

For $E = 0$, the state is the vacuum and the assertion is immediate.

Let $P_K = \mathbf{1}_{\{N \leq K\}}$, $Q_K = I - P_K$, and $q = \text{Tr } \rho Q_K$. The rank of P_K is finite, and $q \leq E/(K+1)$. Factoring off $\rho^{1/2}$ and applying the Hilbert–Schmidt Cauchy–Schwarz inequality bounds each off-diagonal block by $\sqrt{q}$ in trace norm. Hence ρ is uniformly approximated by its finite-dimensional compression; normalizing the latter changes its trace norm by at most q . This proves precompactness. The energy bound is closed by lower semicontinuity of positive spectral expectations, so the set is compact.

For completeness, write $H_2(q) = -q \log q - (1 - q) \log(1 - q)$. Pinching into the two number blocks changes the entropy by a quantity between zero and $H_2(q)$. To see the upper bound directly, split each vector in a pure eigenensemble of ρ into its two projected vectors. Concavity of H_2 bounds the increase of the ensemble-weight entropy by $H_2(q)$; the entropy of a pure-state mixture is at most its weight entropy. The latter fact follows by applying diagonal entropy to the Gram operator of the weighted vectors, which has the mixture's nonzero eigenvalues. The lower bound follows from the same concavity argument for pinching.

The pinched entropy equals

$$H_2(q) + (1 - q)S\left(\frac{P_K \rho P_K}{1 - q}\right) + qS\left(\frac{Q_K \rho Q_K}{q}\right),$$

where zero-mass terms are omitted. The last term is bounded by

$$qn g\left(\frac{E}{qn}\right),$$

which tends to zero uniformly as $q \downarrow 0$. The low-number entropy term is continuous in the finite-dimensional compression. These bounds uniformly approximate $S(\rho)$ by continuous finite-dimensional expressions and prove its continuity under the common energy bound.

Finally, normalized number compressions converge in trace norm. Their energies are at most $E/(1 - q)$, and hence are uniformly bounded for all sufficiently large K . The established continuity proves entropy convergence. $\square$

In particular, entropy grows only logarithmically with energy for fixed n , whereas a positive linear energy penalty is coercive. For $r > 0$, we shall repeatedly use the identity

$$D(\rho \parallel \tau_r) = -S(\rho) + n \log(r + 1) + h(r) \operatorname{Tr} \rho N. \quad (2.4)$$

2.2 Passive mixing

For any finite collection of ports, a real orthogonal rotation of their mode operators, performed identically for every mode index, is implemented by a unitary on their joint Fock space. One way to construct it is to rotate the creation operators acting on the joint vacuum; the commutation relations show that the resulting number vectors again form an orthonormal basis. This unitary preserves combined total number. A retained subsystem therefore has energy no greater than the combined input energy. On each finite total-number sector, the unitary depends continuously on the rotation; consequently the induced channels are trace-norm continuous on every fixed state as the rotation varies.

Proposition 2.3 (Thermal mixing). *Suppose independent ports carry the n -mode states τ_{r_s} , and the retained modes are $\sum_s \sqrt{\lambda_s} d_{s,j}$, where $\lambda_s \geq 0$ and $\sum_s \lambda_s = 1$. Their joint retained state is*

$$\tau_{\sum_s \lambda_s r_s}.$$

Proof. The coherent-state representation of a thermal state is

$$\tau_r = \int_{\mathbb{C}^n} |z\rangle \langle z| \frac{e^{-\|z\|^2/r}}{(\pi r)^n} d^{2n} z.$$

Indeed, inserting the coefficients $\langle \mathbf{k} | z \rangle = e^{-\|z\|^2/2} z^{\mathbf{k}} / \sqrt{\mathbf{k}!}$ and integrating gives the diagonal geometric probabilities in (2.1). Products of coherent states transform by the same linear rotation of their

amplitudes, as follows from their annihilation eigenvector equations or creation-operator expansion. Independent centered circular complex Gaussians with covariances $r_s I$ consequently produce retained covariance $(\sum_s \lambda_s r_s) I$. The displayed representation proves the claim. Zero reference means follow by continuity. $\square$

This thermal representation and its relation to entropy photon number are also the starting point of the original formulation [20, Section II.B]. In the argument below it is only the auxiliary reference states that have this special form.

3 Interpolation of diagonal form comparisons

The proof requires an interpolation principle for quadratic forms. Its hypotheses involve four simple weights. In the later application at the regularized minima, the target weight makes the squared norm of a centered thermal-balance defect equal to the normalized entropy-production pairing plus a centered-energy correction. This is the identity used in the stationarity cancellation. We prove the principle here, including the analytic property on which it depends.

Set

$$f(x) = \frac{x}{\sinh x}, \quad b(x) = x \coth x, \quad f(0) = b(0) = 1. \quad (3.1)$$

Both functions are positive and even on $\mathbb{R}$. On Hilbert spaces $\mathcal{H}_j$, $0 \leq j \leq k$, fix coordinate representations in which all weights below act by scalar multiplication. A positive weight X_j defines the possibly unbounded form

$$Q_{X,j}(Z) = \sum_{\alpha} X_j(\alpha) |Z_{\alpha}|^2;$$

fixed positive coordinate measures can be absorbed into the coordinates. Let $\mathcal{T} \subset \mathcal{H}_0$ be a linear test space, let $L_i : \mathcal{T} \rightarrow \mathcal{H}_i$ be linear, and let $w_i > 0$ for $1 \leq i \leq k$. We say that the family $X = (X_j)_{j=0}^k$ satisfies comparison if

$$Q_{X,0}(Z) \geq \sum_{i=1}^k w_i Q_{X,i}(L_i Z) \quad (Z \in \mathcal{T}). \quad (3.2)$$

All sums on the right are interpreted as nonnegative coordinate sums.

Theorem 3.1 (Four-weight interpolation). *At every coordinate of every space, let $m > 0$ and $x, y \in \mathbb{R}$ be given. Suppose each $Z \in \mathcal{T}$ is supported on finitely many values of the parameter tuple (m, x, y) , and $\mathcal{T}$ is invariant under scalar multiplication by functions of this tuple. If the four families*

$$mf(x)e^x, \quad mf(x)e^{-x}, \quad mf(y)e^y, \quad mf(y)e^{-y} \quad (3.3)$$

satisfy comparison, then so does

$$F = m \frac{f(x)f(y)}{f(x+y)}. \quad (3.4)$$

Finitely many parameter values do not mean finitely many nonzero coordinates. In particular, a centered finite matrix can have an infinite identity tail, all at the same parameter value. The hypothesis permits this situation, which will occur in our application.

3.1 Operations that preserve comparison

The operations below belong to the parallel-sum and operator-mean calculus of Anderson–Duffin and Kubo–Ando [1, 25]. We give direct form proofs, including the limits needed for unbounded weights.

Throughout this subsection, the auxiliary weights are positive functions of the parameter tuple (m, x, y) . Positive linear combinations preserve (3.2). So do pointwise limits of such weights: on a source test there are only finitely many parameter values, so its form converges; on the target spaces Fatou’s lemma gives the required inequality.

Comparisons also persist under the *parallel sum*

$$X : Y = (X^{-1} + Y^{-1})^{-1}. \quad (3.5)$$

Indeed, coordinatewise,

$$(X : Y)|z|^2 = \min_{u+v=z} (X|u|^2 + Y|v|^2), \quad u = \frac{Y}{X+Y}z, \quad v = \frac{X}{X+Y}z.$$

The source minimizers U, V lie in $\mathcal{T}$. Apply the comparisons for X and Y to them, then use $L_i U + L_i V = L_i Z$ and the coordinatewise minimum on each target space. Target minimizers need not belong to any specified test space.

Consequently, if X, Y satisfy comparison, so do their logarithmic mean $\ell(X, Y) = (X - Y)/(\log X - \log Y)$ and $XY/\ell(X, Y)$, with the continuous value $\ell(X, X) = X$. The identities that implement these operations are

$$\frac{1}{\ell(X, Y)} = \int_0^1 \frac{dt}{(1-t)X + tY}, \quad \frac{XY}{\ell(X, Y)} = \int_0^1 \frac{dt}{t/X + (1-t)/Y}. \quad (3.6)$$

For the first identity, a positive quadrature followed by inversion is a finite parallel sum: if $q_j > 0$ and $V_j > 0$, then

$$\left(\sum_{j=1}^M \frac{q_j}{V_j} \right)^{-1} = (V_1/q_1) : \cdots : (V_M/q_M).$$

Here $V_j = (1 - t_j)X + t_j Y$ already satisfies comparison. The second identity is a positive average of weighted parallel sums. Pointwise limits give the integrals; zero coefficients are omitted or obtained by a limit. Thus no rule asserting preservation under multiplication of weights is being used.

Apply these two operations to each opposite-exponent pair in (3.3). Since their logarithmic mean is m , we obtain comparison for the three families

$$m, \quad ms, \quad mt, \quad s = f(x)^2, \quad t = f(y)^2. \quad (3.7)$$

The next step constructs one further weight from these three.

3.2 A Stieltjes representation

The function $x \mapsto f(x)^2$ decreases from 1 to 0 on $[0, \infty)$. Define B on $(0, 1]$ by

$$B(f(x)^2) = b(x). \quad (3.8)$$

The scalar quadratic (3.18), used at the end of the proof, determines the candidate for the additional weight. It pairs the four original weights on $\xi \mp \phi$ and $\xi \pm \phi$ and adds $E|\phi|^2$. For a candidate $E = me > 0$, write $a = b(x) + b(y)$. Its minimum is

$$\frac{m}{2} \left(a - \frac{(x - y)^2}{a + 2e} \right) |\xi|^2.$$

For $x^2 \neq y^2$, the addition formula gives $F/m = (yb(x) + xb(y))/(x + y)$. Equating the minimum with $F|\xi|^2$ therefore requires, within this quadratic construction,

$$(a + 2e)(b(x) - b(y)) = x^2 - y^2, \quad E = \frac{m(t - s)}{2(B(s) - B(t))}, \quad s = f(x)^2, \quad t = f(y)^2,$$

where the last equality uses $b(z)^2 = f(z)^2 + z^2$. The final minimization will verify the value in all cases. It remains to obtain comparison for this candidate from the known weights. Its reciprocal has the integral form, initially for $0 < s, t < 1$,

$$\frac{1}{E} = \frac{2}{m} \int_0^1 -B'(\theta s + (1 - \theta)t) d\theta, \quad s = f(x)^2, \quad t = f(y)^2. \quad (3.9)$$

The lemma below will show that this defines a positive continuous weight also when $s = 1$ or $t = 1$. A Stieltjes representation of $-B'$ turns the reciprocal into a limit of positive combinations of $1/m$ and reciprocals of $\theta(ms) + (1 - \theta)(mt) + rm$, with $r \geq 0$. The parallel-sum argument above will then construct E from the three known weights. We now prove the required representation.

Lemma 3.2. *The function B extends holomorphically to the upper half-plane and across the positive real axis, is real and strictly decreasing there, and satisfies $B(1) = 1$, $B'(1) = -1$. There are a constant $c \geq 0$ and a positive measure ν on $[0, \infty)$ such that*

$$-B'(s) = c + \int_{[0, \infty)} \frac{d\nu(r)}{s + r} \quad (s > 0), \quad \int_{[0, \infty)} \frac{d\nu(r)}{1 + r} < \infty. \quad (3.10)$$

In particular $-B'(s) > 0$ for every $s > 0$.

Proof. For $x > 0$, put $s = f(x)^2 \in (0, 1)$. Differentiating $s = f(x)^2$ and $B(s) = b(x)$ gives

$$\frac{ds}{dx} = \frac{2s(1 - B)}{x}, \quad \frac{db}{dx} = \frac{B - s}{x}.$$

Here $B(s) = x \coth x > 1$, so we may set

$$R(s) = \frac{1 - s}{s(B(s) - 1)}.$$

The derivatives give

$$-B'(s) = \frac{B(s) - s}{2s(B(s) - 1)} = \frac{1}{2} \left(\frac{1}{s} + R(s) \right). \quad (3.11)$$

The term $1/s$ is already a Stieltjes kernel, so it remains to obtain such a representation for R . We first construct the extension of B using

$$s = \left(\frac{u}{\sin u} \right)^2, \quad B(s) = u \cot u. \quad (3.12)$$

For $q > 0$ let $p_*(q) \in (\pi/2, \pi)$ be the unique solution of

$$p_*(q) \tan p_*(q) = -q \tanh q,$$

and put

$$D = \{p + iq : q > 0, 0 < p < p_*(q)\}, \quad \mathcal{Q} = \{z : \operatorname{Re} z > 0, \operatorname{Im} z < 0\}.$$

Existence, uniqueness and continuity of p_* follow because $p \tan p$ increases strictly from $-\infty$ to 0 on $(\pi/2, \pi)$: its derivative is $(p + \sin p \cos p)/\cos^2 p > 0$. Thus D is connected. The map $a(u) = \sin u/u$ takes D into $\mathcal{Q}$, as follows from

$$\begin{aligned}\operatorname{Re} a(p + iq) &= \frac{p \sin p \cosh q + q \cos p \sinh q}{p^2 + q^2} > 0, \\ \operatorname{Im} a(p + iq) &= \frac{p \cos p \sinh q - q \sin p \cosh q}{p^2 + q^2} < 0.\end{aligned}$$

The first sign is precisely the boundary equation when $p > \pi/2$, and is immediate for $p \leq \pi/2$. For the second, when $p < \pi/2$ use $\tan p/p > 1 > \tanh q/q$; for $p \geq \pi/2$ the sign is immediate.

This map is proper into $\mathcal{Q}$. Its finite boundary pieces map to the coordinate axes: $p = 0$ and $q = 0$ map to the positive real axis, and $p = p_*(q)$ to the negative imaginary axis. The limiting corners $u = 0, \pi$ map to $1, 0$, respectively. Moreover,

$$|a(p + iq)| \geq \frac{\sinh q}{\sqrt{\pi^2 + q^2}} \rightarrow \infty \quad (q \rightarrow \infty)$$

uniformly for $0 \leq p \leq \pi$. Thus the inverse image of a compact subset of $\mathcal{Q}$ cannot approach the boundary or infinity. There are no critical points in D , because

$$\operatorname{Im}(u \cot u) = \frac{q \sin(2p) - p \sinh(2q)}{\cosh(2q) - \cos(2p)} < 0, \quad (3.13)$$

whereas $a'(u) = 0$ would imply $u \cot u = 1$. The strict sign follows from $\sin(2p) \leq 2p$ and $\sinh(2q) > 2q$.

For completeness, properness and the inverse function theorem imply that $a(D)$ is both open and relatively closed in $\mathcal{Q}$, hence is all of $\mathcal{Q}$. Each fiber is finite, and local inverse neighborhoods at its points give an evenly covered neighborhood: an additional inverse point arbitrarily close to the chosen image point would, by properness, accumulate at that fiber. Thus a is a covering map. A connected covering of the simply connected quadrant has one sheet: inverse branches continue along paths by successive evenly covered neighborhoods, and homotopies make the continuations independent of the path. Therefore $a : D \rightarrow \mathcal{Q}$ has a holomorphic inverse.

For $\operatorname{Im} s > 0$, take the branch of $s^{-1/2}$ in $\mathcal{Q}$ and define $u = a^{-1}(s^{-1/2})$. Equations (3.12) and (3.13) give a holomorphic B with $\operatorname{Im} B < 0$. To check its continuation at $s_0 > 0$, the preceding bound on $|a|$ bounds q as $s \rightarrow s_0$ from above. Every cluster point of u lies on $p = 0$ or $q = 0$, since the curved boundary has purely imaginary image. On these pieces $\sinh q/q$ increases strictly from 1 to ∞ , and $\sin p/p$ decreases strictly from 1 to 0 for $0 < p < \pi$. These observations identify a unique boundary limit: $u = iq$ if $0 < s_0 < 1$, and $u = p \in (0, \pi)$ if $s_0 > 1$. The derivatives there are nonzero, so local analytic inversion gives continuation across s_0 . At $s_0 = 1$ use u^2 as parameter:

$$\left(\frac{u}{\sin u}\right)^2 = 1 + \frac{u^2}{3} + O(u^4), \quad u \cot u = 1 - \frac{u^2}{3} + O(u^4).$$

This gives $B(1) = 1$, $B'(1) = -1$. The two real parametrizations also show that B is strictly decreasing on $(0, \infty)$ and agrees with (3.8).

We next recall the classical Herglotz representation [22], with a positive-harmonic proof in the form needed below. If P is holomorphic with $\operatorname{Im} P \geq 0$ in the upper half-plane and extends holomorphically with real values across $(0, \infty)$, then

$$P(s) = a_0 + \beta s + \int_{(-\infty, 0]} \left(\frac{1}{r-s} - \frac{r}{1+r^2} \right) d\mu(r), \quad \beta \geq 0, \quad (3.14)$$

where $a_0 \in \mathbb{R}$, $\mu \geq 0$ and $\int (1+r^2)^{-1} d\mu(r) < \infty$. Indeed, under $s = i(1+z)/(1-z)$ its imaginary part becomes a nonnegative harmonic function v on the disk. The measures $v(\varrho e^{i\theta}) d\theta/(2\pi)$ have the fixed mass $v(0)$. A weak limit as $\varrho \uparrow 1$, together with the Poisson formula on each smaller disk, represents v by a positive boundary measure. The atom at $z = 1$ contributes $\beta \operatorname{Im} s$; at the other boundary points the Poisson kernel becomes $(1+r^2) \operatorname{Im} s/|r-s|^2$. Incorporate the factor $1+r^2$ into $d\mu(r)$. The measure has no mass on the arcs corresponding to $r > 0$, because v tends uniformly to zero on their compact subarcs, by the assumed continuation. The analytic integral in (3.14) converges locally uniformly; its imaginary part is the one just obtained. The remaining holomorphic function has zero imaginary part and is a real constant. This proves (3.14).

Apply (3.14) to $P = -B$ and subtract the value at 1. The function

$$G(s) = \frac{B(s) - 1}{1 - s} = \beta + \int_{(-\infty, 0]} \frac{d\mu(r)}{(1-r)(s-r)} \quad (3.15)$$

has a removable value $G(1) = 1$. It is positive on the positive real axis, and $H(s) = sG(s)$ has nonnegative imaginary part in the upper half-plane: this follows termwise from $\beta \geq 0$ and $\operatorname{Im}(s/(s-r)) \geq 0$ for $r \leq 0$. The function H has no zero there. Otherwise the nonnegative harmonic function $\operatorname{Im} H$ would attain zero inside, and the minimum principle would force H to be a real constant; its positive real-axis values rule out the constant zero. Thus

$$R(s) = \frac{1}{sG(s)}$$

continues the real function R above holomorphically to the upper half-plane and across $(0, \infty)$, is positive on that interval, and satisfies $\operatorname{Im}(-R) \geq 0$.

Apply (3.14) once more, now to $-R$, with coefficients $\tilde{\beta}, \tilde{\mu}$. For real $s > 1$ it gives

$$\begin{aligned} 0 \leq R(1) - R(s) &= \tilde{\beta}(s-1) + \int_{(-\infty, 0]} \left(\frac{1}{1-r} - \frac{1}{s-r} \right) d\tilde{\mu}(r) \\ &\leq R(1). \end{aligned}$$

Consequently $\tilde{\beta} = 0$, and monotone convergence gives $\int (1-r)^{-1} d\tilde{\mu}(r) \leq R(1)$. We may therefore write

$$R(s) = c_0 + \int_{(-\infty, 0]} \frac{d\tilde{\mu}(r)}{s-r}, \quad c_0 = R(1) - \int_{(-\infty, 0]} \frac{d\tilde{\mu}(r)}{1-r} \geq 0. \quad (3.16)$$

Initially this identity follows on the positive axis; local uniform convergence and analytic continuation give it in the upper half-plane as well.

The holomorphic functions $-B'$ and $\frac{1}{2}(1/s + R(s))$ agree on $0 < s < 1$ by (3.11). The identity theorem extends their equality to the upper half-plane and the positive real axis. Equation (3.16), after replacing r by $-r$, now proves (3.10); the term $1/(2s)$ contributes an atom of mass $1/2$ at zero and ensures strict positivity. $\square$

3.3 The additional weight and the final minimization

The lemma extends the same integral definition (3.9) continuously and positively to all $s, t > 0$, including the values used in (3.7). The fundamental theorem of calculus gives

$$E = \begin{cases} \frac{m(t-s)}{2(B(s) - B(t))}, & s \neq t, \\ -\frac{1}{2B'(s)}, & s = t. \end{cases} \quad (3.17)$$

Lemma 3.2 shows that E is obtainable by the comparison-preserving operations already established. Indeed, after substitution of (3.10), every reciprocal term is of the form

$$\frac{1}{m(\theta s + (1 - \theta)t + r)} = \frac{1}{\theta(ms) + (1 - \theta)(mt) + rm},$$

and the constant term is a multiple of $1/m$. Positive finite quadratures followed by inversion are parallel sums of positive multiples of these positive linear combinations of m, ms, mt . Truncating the r -integral and refining partitions gives pointwise convergence for every $s, t > 0$: on each bounded interval the integrands are continuous, and the tail is controlled by $\int (1 + r)^{-1} d\nu(r) < \infty$. The source finite-parameter hypothesis and target Fatou argument therefore prove comparison for E .

We complete the proof of Theorem 3.1 by an exact scalar minimization. For $\xi, \phi \in \mathbb{C}$ put

$$\mathcal{Q}(\xi, \phi) = \frac{1}{4} \sum_{\pm} m f(x) e^{\pm x} |\xi \mp \phi|^2 + \frac{1}{4} \sum_{\pm} m f(y) e^{\pm y} |\xi \pm \phi|^2 + E |\phi|^2. \quad (3.18)$$

Write $a = b(x) + b(y)$ and $e = E/m$. Since $f(x)e^{\pm x} = b(x) \pm x$, expansion gives

$$\mathcal{Q}(\xi, \phi) = \frac{ma}{2} |\xi|^2 + m \left(\frac{a}{2} + e \right) |\phi|^2 - m(x - y) \operatorname{Re}(\bar{\xi}\phi).$$

The unique minimizer and its value are

$$\phi = \frac{x - y}{a + 2e} \xi, \quad \min_{\phi} \mathcal{Q}(\xi, \phi) = \frac{m}{2} \left(a - \frac{(x - y)^2}{a + 2e} \right) |\xi|^2. \quad (3.19)$$

If $x^2 \neq y^2$, the identity $b(z)^2 = f(z)^2 + z^2$ and (3.17) give

$$(a + 2e)(b(x) - b(y)) = x^2 - y^2.$$

Hence the minimum in (3.19) is

$$m \frac{yb(x) + xb(y)}{x + y} |\xi|^2 = m \frac{f(x)f(y)}{f(x + y)} |\xi|^2, \quad (3.20)$$

where the second equality is the addition formula for $\sinh$, with continuous values when x or y is zero.

No singular minimization is hidden when $x^2 = y^2$. If $y = x$, the minimizer is $\phi = 0$ and the value is $mb(x)|\xi|^2 = mf(x)^2|\xi|^2/f(2x)$. If $y = -x \neq 0$, write $s = f(x)^2$ and $b = b(x)$. Equations (3.17) and (3.11) yield

$$e = \frac{s(b - 1)}{b - s}, \quad b + e = \frac{x^2}{b - s}.$$

Thus (3.19) gives $m(b - x^2/(b + e))|\xi|^2 = ms|\xi|^2$, as required. At $x = y = 0$, we have $E = m/2$, $\phi = 0$, and the value is $m|\xi|^2$. This proves (3.20) in every case.

For a source test Z , let Φ be its coordinatewise minimizer in (3.19). It belongs to $\mathcal{T}$ by the invariance assumption. Apply the four assumed comparisons to $Z \mp \Phi$ and $Z \pm \Phi$, and the comparison for E to Φ . After summing, the source side is $Q_{F,0}(Z)$ by (3.20), whereas the target side is

$$\sum_i w_i \sum_{\alpha} \mathcal{Q}_i((L_i Z)_{\alpha}, (L_i \Phi)_{\alpha}) \geq \sum_i w_i Q_{F,i}(L_i Z).$$

This is the claimed comparison, and proves Theorem 3.1.

4 A regularized minimum and its Hessian

We argue by contradiction. This section constructs a strictly negative minimum with enough regularity to support the later operator identities. It then extracts an entropy-Hessian estimate at that minimum.

4.1 The auxiliary ports and the functional

Suppose Theorem 1.1 fails. The endpoint transmissions give equality, so $0 < \eta < 1$. By Lemma 2.2, normalized number cutoffs of the inputs converge with bounded energies and convergent entropies, as do their outputs. Thus there is a fixed strictly violating pair (ρ_1^w, ρ_2^w) with finite number support. Write $a_i = g^{-1}(S(\rho_i^w)/n)$. At least one a_i is positive.

Besides the two variable ports 1, 2, introduce a fixed thermal port D , in state $\rho_D = \tau_{r_D}$ with $r_D > 0$. Its mixing weight and those of the variable inputs are

$$0 < \lambda_D < 1, \quad \lambda_1 = (1 - \lambda_D)\eta, \quad \lambda_2 = (1 - \lambda_D)(1 - \eta).$$

The retained port, denoted by 0, has annihilators $d_{0,j} = \sum_{s=1,2,D} \sqrt{\lambda_s} d_{s,j}$ before reduction. For positive reference parameters r_1, r_2 , set

$$r_0 = \lambda_1 r_1 + \lambda_2 r_2 + \lambda_D r_D, \quad h_s = h(r_s) \quad (s = 0, 1, 2, D).$$

Let γ be the retained state obtained from $\rho_1 \otimes \rho_2 \otimes \rho_D$. We also replace a small fraction of that output by its reference thermal:

$$\rho_0 = (1 - \kappa)\gamma + \kappa\tau_{r_0}, \quad 0 < \kappa < 1, \quad w_i = (1 - \kappa)\lambda_i \quad (i = 1, 2).$$

Unless otherwise indicated, sums indexed by i run over 1, 2. Since $h(r) = g'(r)$, the quotient $(S(\rho) - ng(r))/h(r)$ expresses the entropy deviation in the linearized scale of total photon number at the thermal reference. For $\varepsilon, \zeta > 0$ we minimize, over input states of finite fourth moment, the functional

$$\begin{aligned} J_\zeta(\rho_1, \rho_2) &= \frac{S(\rho_0) - ng(r_0)}{h_0} - \sum_i w_i \frac{S(\rho_i) - ng(r_i)}{h_i} \\ &\quad + \varepsilon \sum_i \frac{w_i}{h_i} D(\rho_i \| \tau_{r_i}) + \zeta \sum_i \text{Tr } \rho_i W_i^4. \end{aligned} \tag{4.1}$$

The normalization of the relative-entropy penalty also matches the closing stationarity calculation. The thermal entropy formula and (2.4) give

$$\frac{D(\rho \| \tau_r)}{h(r)} = -\frac{S(\rho) - ng(r)}{h(r)} + (\text{Tr } \rho N - nr).$$

Thus the input terms pair an entropy coefficient $1 + \varepsilon$ with an energy coefficient ε . Their centered-energy coefficients differ by one in stationarity, matching the output centered-energy relation while retaining the strict defect and mean terms.

The parameters are chosen in the following order. First choose $r_i > 0$ equal or sufficiently close to a_i . At $\lambda_D = \kappa = \varepsilon = \zeta = 0$, the value at the fixed witness approaches

$$\frac{S(\rho_C^w) - ng(\eta a_1 + (1 - \eta)a_2)}{h(\eta a_1 + (1 - \eta)a_2)} < 0.$$

The input differences vanish in this limit. If $a_i = 0$, this follows from $g(r)/h(r) \rightarrow 0$ as $r \downarrow 0$; the output denominator has a strictly positive, finite limit. Fix $r_D > 0$ and then choose $\lambda_D > 0$ sufficiently small to retain negativity. Indeed, the additional mixing can be performed after the original beam splitter; as $\lambda_D \downarrow 0$, its unitary converges strongly on each total number sector. The resulting states converge in trace norm with bounded energy, and their entropies converge by Lemma 2.2.

Next choose $\varepsilon > 0$ small enough to retain negativity and to have

$$(1 + \varepsilon) \sum_i \lambda_i r_i \leq r_0, \quad (1 + \varepsilon) \sum_i \lambda_i (r_i + 1) \leq r_0 + 1. \quad (4.2)$$

At $\varepsilon = 0$ the gaps are respectively $\lambda_D r_D$ and $\lambda_D (r_D + 1)$, both positive. Finally take $\kappa > 0$ sufficiently small. The additional bound on κ used in Section 5 depends only on the parameters already fixed, so it is compatible with this choice. The witness has finite fourth moments and finite thermal relative entropies. Consequently there are constants $c > 0$ and $\zeta_0 > 0$ such that

$$J_\zeta(\rho_1^w, \rho_2^w) \leq -c \quad (0 < \zeta \leq \zeta_0). \quad (4.3)$$

All parameters except ζ remain fixed from now on.

4.2 A Gibbs regularity lemma

The fourth-moment penalty yields more regularity than its definition alone suggests. We record the needed form argument explicitly.

Lemma 4.1. *Let $W = 1 + N$ on the n -mode Fock space. Suppose $k > 0$ and B is a bounded positive operator. The positive form*

$$K = kW^4 + W^{1/2}BW^{1/2}, \quad \mathcal{D}(K^{1/2}) = \mathcal{D}(W^2),$$

defines a positive self-adjoint operator with compact inverse. Its normalized Gibbs state $\omega = e^{-K} / \text{Tr } e^{-K}$ is faithful and satisfies $\text{Tr } \omega W^7 < \infty$. It uniquely minimizes $\text{Tr } \sigma K - S(\sigma)$ among states with finite fourth moment.

Proof. Put $C = kI + W^{-3/2}BW^{-3/2}$. The form is $\|C^{1/2}W^2\phi\|^2$ on $\mathcal{D}(W^2)$, and is closed because $kI \leq C \leq \|C\|I$. The operator

$$U = W^{-2}C^{-1}W^{-2}$$

is positive, compact, and injective. Choose a complete orthonormal eigenbasis ψ_j with $U\psi_j = u_j^{-1}\psi_j$ and $u_j > 0$. Every ψ_j belongs to $\mathcal{D}(W^2)$ and satisfies the form eigenvalue equation for K with eigenvalue u_j . For completeness, the vectors $\sqrt{u_j}C^{-1/2}W^{-2}\psi_j$ are orthonormal and complete: orthonormality follows from U , and completeness from the dense range of $C^{-1/2}W^{-2}$. Parseval's identity therefore gives

$$\|C^{1/2}W^2\phi\|^2 = \sum_j u_j |\langle \psi_j, \phi \rangle|^2 \quad (\phi \in \mathcal{D}(W^2)).$$

Closedness and finite spectral sums show that the domain of this diagonal form is exactly $\mathcal{D}(W^2)$. This constructs $K = U^{-1}$.

The bound $U \leq k^{-1}W^{-4}$ gives

$$\#\{j : u_j \leq a\} \leq \text{rank } \mathbf{1}_{\{W^4 \leq a/k\}},$$

by comparing dimensions with the complementary number subspace. The right side grows polynomially in a . Also $\|W^2\psi_j\|^2 \leq u_j/k$. Test the form eigenvalue equation against each number vector and multiply that coordinate equation by $W^{-1/2}$. It gives

$$kW^{7/2}\psi_j = u_jW^{-1/2}\psi_j - BW^{1/2}\psi_j.$$

The right side is square summable, so $\psi_j \in \mathcal{D}(W^{7/2})$, with the explicit bound

$$\|W^{7/2}\psi_j\| \leq k^{-1}(u_j + \|B\|\sqrt{u_j/k}).$$

Polynomial eigenvalue counting and the exponential weights e^{-u_j} now prove both $0 < \text{Tr } e^{-K} < \infty$ and $\text{Tr}(e^{-K}W^7) < \infty$. Finally, for a state σ with finite fourth moment, all terms in

$$\text{Tr } \sigma K - S(\sigma) = D(\sigma\|\omega) - \log \text{Tr } e^{-K}$$

are finite. Nonnegativity and the equality case of relative entropy give the asserted unique minimum. $\square$

4.3 Attainment and the exact logarithm identity

Proposition 4.2. *For every $0 < \zeta \leq \zeta_0$, the functional J_ζ attains a minimum of value at most $-c$. The input energies at these minima are bounded uniformly in ζ . At each minimum all four states ρ_s , $s = 0, 1, 2, D$, are faithful, have finite seventh moment, and satisfy $0 \leq H_s := -\log \rho_s \leq C_s W_s^4$ as forms. In fact $H_0 \leq CW_0$. The inputs obey the exact identities*

$$\frac{(1+\varepsilon)w_i}{h_i}H_i = \frac{1-\kappa}{h_0}T_iH_0 + \varepsilon w_iN_i + \zeta W_i^4 + c_iI, \quad i = 1, 2, \quad (4.4)$$

where T_i is the output-observable slice with the other two inputs fixed, and c_i is a real scalar. The moment and logarithm bounds need not be uniform as $\zeta \downarrow 0$.

Proof. Write $E_i = \text{Tr } \rho_i N_i$. Expanding the thermal relative entropy expresses J_ζ , up to constants depending only on the fixed parameters, as

$$\frac{S(\rho_0)}{h_0} - \sum_i \frac{(1+\varepsilon)w_i}{h_i}S(\rho_i) + \varepsilon \sum_i w_i E_i + \zeta \sum_i \text{Tr } \rho_i W_i^4. \quad (4.5)$$

The first term is nonnegative. The bound $S(\rho_i) \leq ng(E_i/n) = O(\log(1+E_i))$ shows that the positive linear energy terms dominate every negative entropy term. Thus bounded upper sublevels have bounded input energies, uniformly in $0 < \zeta \leq \zeta_0$. For fixed $\zeta > 0$ their fourth moments are bounded as well. Take a minimizing sequence in the sublevel supplied by (4.3). Lemma 2.2 gives a trace-norm convergent subsequence of each input. The product and channel maps are trace-norm continuous, and output energies are bounded by the combined input energy plus the fixed replacement energy. Entropy is continuous along this subsequence. The energy and fourth-moment expectations are lower semicontinuous, by bounded spectral truncation. Equation (4.5) therefore gives an admissible minimum, and the same sublevel estimate gives its uniform energy bound.

Fix such a minimum. The replacement gives $\rho_0 \geq \kappa\tau_{r_0}$. The resolvent representation of the logarithm, together with inverse order for positive operators, implies

$$0 \leq H_0 \leq -\log(\kappa\tau_{r_0}) \leq CW_0$$

as positive forms. To define T_i , pull an output observable back through the passive unitary and take its partial expectation in the states of the other two ports. For bounded Z , this is a bounded unital positive slice satisfying

$$\mathrm{Tr} \rho_i T_i Z = \mathrm{Tr} \gamma Z. \quad (4.6)$$

For H_0 , use increasing bounded spectral truncations to define the positive form $T_i H_0$. Since the retained number is bounded by total number before reduction, slicing $H_0 \leq C W_0$ yields

$$0 \leq T_i H_0 \leq C_i W_i.$$

In particular, its expectation is the output cross entropy for every varied input of finite energy.

Vary only input i and replace the output entropy in J_ζ by the cross entropy $\mathrm{Tr} \tilde{\rho}_0 H_0$. Nonnegativity of relative entropy makes the resulting functional $\bar{J}_i$ a majorant, with equality at the chosen minimum. Thus

$$\bar{J}_i(\tilde{\rho}_i) \geq J_\zeta(\tilde{\rho}_i) \geq J_\zeta(\rho_i) = \bar{J}_i(\rho_i).$$

After division by its entropy coefficient $(1+\varepsilon)w_i/h_i$, this majorant is $\mathrm{Tr} \tilde{\rho}_i K_i - S(\tilde{\rho}_i)$ plus a constant, where

$$K_i = \frac{h_i}{(1+\varepsilon)w_i} \left(\frac{1-\kappa}{h_0} T_i H_0 + \varepsilon w_i N_i + \zeta W_i^4 \right). \quad (4.7)$$

This is a form $k W_i^4 + R_i$ with $k > 0$ and $0 \leq R_i \leq C_i W_i$. Hence $R_i = W_i^{1/2} B_i W_i^{1/2}$ for a bounded positive B_i . Lemma 4.1 identifies the input uniquely as

$$\rho_i = \frac{e^{-K_i}}{\mathrm{Tr} e^{-K_i}}.$$

It is faithful, has finite seventh moment, and satisfies $H_i = K_i + \log \mathrm{Tr} e^{-K_i}$, proving (4.4) and the claimed input logarithm bounds.

The fixed thermal port has every number moment. Total-number preservation and $(x_1 + x_2 + x_D)^7 \leq 3^6(x_1^7 + x_2^7 + x_D^7)$ for nonnegative numbers give finite seventh moment of the raw output; the thermal replacement preserves this property. This proves all remaining assertions. $\square$

4.4 The component entropy-Hessian estimate

Work at one of the minima in Proposition 4.2. For each label s , fix an eigenbasis $\rho_s = \mathrm{diag}(p_{s,l})$, with all $p_{s,l} > 0$. These orthonormal eigenbases generally differ from the number bases used to define the energy, and they depend on the minimizing states. With $\ell(a, b) = (a - b)/(\log a - \log b)$ and $\ell(a, a) = a$, define

$$\|Z\|_s^2 = \sum_{l,r} h_s \ell(p_{s,l}, p_{s,r}) |Z_{lr}|^2, \quad v_s(l, r) = \frac{\log(p_{s,l}/p_{s,r})}{h_s}. \quad (4.8)$$

Up to the factor h_s , this is the Bogoliubov inner product on observables. Its dual form on state perturbations is the Bogoliubov–Kubo–Mori metric, obtained from the Hessian of relative entropy [27, 26]. We establish the needed differentiation formulas on finite eigenblocks and then use density; no bounded inverse for the full state is assumed.

These weighted matrix spaces are Hilbert spaces, with inner product conjugate-linear in the first variable. Bounded observables belong to them because the logarithmic mean is at most the arithmetic mean. The centered subspace is the orthogonal complement of I ; the orthogonal projection subtracts $\langle Z \rangle_s I$, where $\langle Z \rangle_s = \mathrm{Tr} \rho_s Z$. Set

$$L_i Z = \frac{T_i Z - \langle T_i Z \rangle_i I}{\sqrt{\lambda_i}}. \quad (4.9)$$

Initially the centered output tests are finite eigenblock matrices with their expectation times I subtracted. They form a dense subspace. Such a test can have an infinite identity tail, but has only finitely many frequencies v_0 . Multiplication by any function of v_0 preserves this class and its centering: every diagonal entry, including that tail, has frequency zero.

Proposition 4.3. *For $i = 1, 2$, the slice map on the centered output space satisfies*

$$\|L_i Z\|_i^2 \leq \frac{1+\varepsilon}{1-\kappa} \|Z\|_0^2. \quad (4.10)$$

It consequently extends continuously to the completed centered spaces. For bounded observables this extension agrees with (4.9).

Proof. For a single varied input of finite fourth moment, denote the resulting modified output by $\tilde{\rho}_0$. The logarithm identity (4.4) cancels all linear terms in the objective difference, giving exactly

$$J_\zeta(\tilde{\rho}_i) - J_\zeta(\rho_i) = \frac{(1+\varepsilon)w_i}{h_i} D(\tilde{\rho}_i\|\rho_i) - \frac{1}{h_0} D(\tilde{\rho}_0\|\rho_0) \geq 0. \quad (4.11)$$

All cross entropies are finite by the fourth-moment assumptions and the bounds on H_i, H_0 .

Let A be a traceless Hermitian matrix on a finite input eigenblock. Then $\rho_i + tA$ is a state for all sufficiently small positive and negative t . Its fourth moment is finite: each selected eigenvector has finite seventh moment by the construction in Lemma 4.1. Finite-dimensional differentiation on that block gives

$$D(\rho_i + tA\|\rho_i) = \frac{t^2}{2} \mathcal{Q}_i(A) + o(t^2), \quad \mathcal{Q}_i(A) = \sum_{l,r} \frac{|A_{lr}|^2}{\ell(p_{i,l}, p_{i,r})}.$$

For example, the derivative of the logarithm is the integral of $(\rho_i + uI)^{-1} A (\rho_i + uI)^{-1}$ over $u > 0$ on this block.

To extract a dual lower bound without restricting the actual output perturbation to a finite eigenblock, we test the Gibbs variational formula with a centered output observable. Let Z be a Hermitian centered output test as above. The Gibbs inequality gives

$$D(\tilde{\rho}_0\|\rho_0) \geq t \operatorname{Tr} \tilde{\rho}_0 Z - \log \operatorname{Tr} \exp(\log \rho_0 + tZ).$$

The logarithm of this partition function has first derivative zero and second derivative $\|Z\|_0^2/h_0$ at zero. This follows from finite-block exponential differentiation: before centering, only a finite eigenblock changes, and centering merely multiplies the exponential by a scalar. Writing $Z_t = \operatorname{Tr} e^{\log \rho_0 + tZ}$ and $\sigma_t = Z_t^{-1} e^{\log \rho_0 + tZ}$, the same block decomposition gives $0 < Z_t < \infty$ and makes σ_t faithful. Since $-\log \sigma_t = H_0 - tZ + \log Z_t$, the finite energy of $\tilde{\rho}_0$, the bound $H_0 \leq CW_0$, and boundedness of Z give finite entropy and finite cross entropy against σ_t . Lemma 2.1 therefore gives the displayed lower bound. Since the output perturbation induced by tA is $(1-\kappa)t$ times the raw channel perturbation, the lower bound has expansion

$$t^2 \left((1-\kappa) \operatorname{Tr} A T_i Z - \frac{\|Z\|_0^2}{2h_0} \right) + o(t^2).$$

Put $\alpha = (1+\varepsilon)w_i h_0/h_i$. Comparing with (4.11) yields

$$2(1-\kappa) \operatorname{Tr} A T_i Z - \alpha \mathcal{Q}_i(A) \leq \frac{\|Z\|_0^2}{h_0}. \quad (4.12)$$

We can optimize the left side over the completed traceless Hermitian space for $\mathcal{Q}_i$. Indeed, trace is continuous in this norm, since

$$|\operatorname{Tr} A|^2 \leq \left(\sum_l p_{i,l} \right) \sum_l \frac{|A_{ll}|^2}{p_{i,l}} \leq \mathcal{Q}_i(A).$$

Finite eigenblock truncation followed by correcting the trace at one fixed diagonal entry proves density in the trace-zero subspace. The constrained dual of $\mathcal{Q}_i$ is the logarithmic-mean norm of the observable after subtracting its ρ_i expectation. Thus the supremum in (4.12) equals

$$\frac{(1-\kappa)^2}{\alpha h_i} \|T_i Z - \langle T_i Z \rangle_i I\|_i^2.$$

It follows that

$$(1-\kappa)^2 \|T_i Z - \langle T_i Z \rangle_i I\|_i^2 \leq (1+\varepsilon) w_i \|Z\|_0^2.$$

Dividing by λ_i gives (4.10). The estimate extends to complex tests by writing real and imaginary Hermitian parts: symmetry of the logarithmic-mean weights and preservation of adjoints make each squared norm the sum of the corresponding two squared norms.

Density now gives the continuous extension. To identify it on a bounded Z , approximate by $P_m Z P_m$, where P_m are increasing finite spectral projections of ρ_0 , and center each approximant. They converge in both output arithmetic norms and hence in the logarithmic-mean norm. The slice is completely positive and unital, so its Schwarz inequality bounds the corresponding input arithmetic norms by the raw output norms. Those in turn are at most $(1-\kappa)^{-1}$ times the modified output norms, because $\rho_0 \geq (1-\kappa)\gamma$. The sliced approximants therefore converge to the bounded slice in (4.9), as required. $\square$

5 The metric adapted to thermal relaxation

Fix a minimizing pair from Proposition 4.2. All comparisons in this section concern these states. We use the centered observable spaces, modular frequencies v_s , and maps L_i introduced in Section 4; sums over i run over 1, 2. For a positive scalar weight $X_s(v_s)$, write

$$\|Z\|_{X_s,s}^2 = \langle Z, X_s Z \rangle_s.$$

We say that a family of weights satisfies comparison when

$$\sum_i w_i \|L_i Z\|_{X_{i,i}}^2 \leq \|Z\|_{X_{0,0}}^2. \quad (5.1)$$

Initially, Z is a finite eigenblock matrix centered by a scalar multiple of the identity. Such tests have finitely many frequencies, including zero; they need not have finite matrix support after centering.

5.1 Four elementary comparisons

Define, with continuous values at zero,

$$b_s(v) = \frac{v}{2} \coth \frac{h_s v}{2}, \quad R_{s,\pm}(v) = b_s(v) \pm \frac{v}{2}, \quad b_s(0) = \frac{1}{h_s}.$$

If $\rho_s = \operatorname{diag}(p_l)$, then

$$h_s \ell(p_l, p_r) R_{s,+}(v_s(l, r)) = p_l, \quad h_s \ell(p_l, p_r) R_{s,-}(v_s(l, r)) = p_r. \quad (5.2)$$

Thus the corresponding squared norms of a centered observable are $\text{Tr } \rho_s Z Z^\dagger$ and $\text{Tr } \rho_s Z^\dagger Z$.

Both families R_+ and R_- satisfy comparison. To see this, embed Z as an observable $\tilde{Z}$ on the three input ports and equip the joint observable space with either arithmetic inner product. The centered one-port subspaces are mutually orthogonal because the joint state is $\rho_1 \otimes \rho_2 \otimes \rho_D$. The centered slice $T_i Z - \text{Tr } \gamma Z I$ is the orthogonal projection onto the i th such subspace: its pairing with a bounded one-port observable is the defining partial-expectation identity. Consequently the sum of the two slice variances is at most the corresponding variance in γ . The variance in $\rho_0 = (1 - \kappa)\gamma + \kappa\tau_{r_0}$ is at least $(1 - \kappa)$ times that variance in γ . Since $w_i/\lambda_i = 1 - \kappa$, these statements give (5.1) for both multiplication orders. Only independence across ports has been used.

The two constant families r_s and $r_s + 1$ also satisfy comparison. Indeed, Proposition 4.3 and (4.2) give

$$\sum_i w_i r_i \|L_i Z\|_i^2 \leq (1 + \varepsilon) \sum_i \lambda_i r_i \|Z\|_0^2 \leq r_0 \|Z\|_0^2,$$

and the same argument applies with $r_s + 1$.

Proposition 5.1 (Interpolated comparison). *The positive weights*

$$F_s(v) = \frac{1}{h_s} \frac{f(h_s v/2) f(-h_s/2)}{f(h_s(v-1)/2)} = \frac{b_s(v) - (r_s + \frac{1}{2})v}{1 - v} \quad (5.3)$$

satisfy (5.1). The quotient at $v = 1$ is understood continuously. For each fixed $r_s > 0$ there are constants $0 < c_s \leq C_s < \infty$, independent of the minimizing states, such that $c_s \leq F_s(v) \leq C_s$ for every $v \in \mathbb{R}$.

Proof. Apply Theorem 3.1 with

$$m = 1/h_s, \quad x = h_s v_s/2, \quad y = -h_s/2.$$

The ambient coordinate spaces in Theorem 3.1 are the full weighted matrix spaces; its source test space consists of the centered finite-eigenblock tests described above. Multipliers depending on the frequency preserve that test space, because every diagonal entry has frequency zero. The first two weights are $R_{s,+}, R_{s,-}$; the other two are $r_s, r_s + 1$. The conclusion is the first expression in (5.3). The second follows from the hyperbolic addition formula and $\coth(h_s/2) = 2r_s + 1$. The first expression is continuous and strictly positive, while the second gives the limits r_s at $+\infty$ and $r_s + 1$ at $-\infty$. This proves both bounds. $\square$

The thermal weight has a useful logarithmic-mean interpretation:

$$\ell(p_l, p_r) F_s(v_s(l, r)) = \ell(r_s p_l, (r_s + 1) p_r). \quad (5.4)$$

Indeed the logarithmic difference on the right is $h_s(v_s(l, r) - 1)$; substitution in (5.3) proves the identity, including its continuous values. Up to a fixed normalization, this is the frequency-dependent logarithmic mean used in the detailed-balance entropy geometry of Carlen and Maas [5, Definition 5.7 and Lemma 5.8]. Proposition 5.1 establishes the particular comparison needed at our regularized minima.

Write $\|Z\|_{F_s} = \|Z\|_{F_{s,s}}$, and let $\|q\|_{F_{s,*}}$ denote the dual norm of a linear functional on the centered space. For a list of n functionals, its squared norm is the sum of the n squared dual norms. Density and Proposition 4.3 extend Proposition 5.1 to the completed centered spaces. On bounded observables the extension of L_i is the actual centered slice. In fact, bounded spectral compressions of Z converge strongly with their adjoints, as do their slices. Dominated convergence in either arithmetic norm gives convergence in the logarithmic-mean norm, since $\ell(a, b) \leq (a + b)/2$. This also explains why the comparison applies to fixed number-basis tests later, although the minimizing eigenbases vary.

5.2 The thermal defect functionals

Let $d_{s,j}$ be the annihilator of mode j on port s , and put

$$\mu_s = (\text{Tr } \rho_s d_{s,j})_{j=1}^n, \quad d'_{s,j} = d_{s,j} - (\mu_s)_j I, \quad e'_s = \text{Tr } \rho_s N_s - |\mu_s|^2.$$

For bounded Z , define

$$q_{s,j}(Z) = (r_s + 1) \text{Tr } \rho_s (d'_{s,j})^\dagger Z - r_s \text{Tr } \rho_s Z (d'_{s,j})^\dagger. \quad (5.5)$$

A coherent displacement shifts $d_{s,j}$ and $(\mu_s)_j$ by the same constant, so these centered functionals also vanish on displaced thermal states. The final argument must control μ_s separately to recover the fixed, undisplaced thermal reference. These are well-defined weak trace pairings and vanish on I . For example, $d\rho = (d\rho^{1/2})\rho^{1/2}$ is trace class by Hilbert–Schmidt factorization whenever $\text{Tr } \rho N < \infty$; the same holds for $d^\dagger\rho$, ρd , and $\rho d^\dagger$. Thus no domain invariance for an arbitrary bounded Z is being assumed.

We record the summability needed for both the metric estimate and the subsequent generator calculation.

Lemma 5.2 (Dual norm and logarithmic sums). *At each minimizing state, q_s has finite dual norm. Set*

$$K_{s,lr} = (r_s + 1)p_r - r_s p_l.$$

Then

$$\|q_s\|_{F_s,*}^2 = \sum_{j,l,r} (1 - v_s(l, r)) K_{s,lr} |(d'_{s,j})_{lr}|^2. \quad (5.6)$$

The sums obtained by replacing the coefficient on the right by $|K_{s,lr}|$ or $|v_s(l, r) K_{s,lr}|$ are finite.

Proof. Suppress the label s and write $H = -\log \rho$. By Proposition 4.2, $\text{Tr } \rho W^7 < \infty$ and $0 \leq H \leq CW^4$ as forms. Every eigenvector ψ_r of ρ belongs to $D(W^{7/2})$, since $p_r > 0$. Number shifts give $\|W^2 d' \psi\| + \|W^2 (d')^\dagger \psi\| \leq C' \|W^{5/2} \psi\|$. Consequently the crossed logarithmic sums, such as

$$\sum_{l,r} p_r (-\log p_l) |d'_{lr}|^2 = \sum_r p_r \langle d' \psi_r, H d' \psi_r \rangle,$$

are bounded by a constant times $\text{Tr } \rho W^5$. For the uncrossed sums, use

$$-\log p_r \leq C \langle \psi_r, W^4 \psi_r \rangle, \quad \langle W^4 \rangle_{\psi_r} \langle W \rangle_{\psi_r} \leq \langle W^5 \rangle_{\psi_r}.$$

The last inequality is the elementary nonnegative covariance inequality for the increasing functions t^4, t of the same positive random variable. It bounds $(-\log p_r) \|d' \psi_r\|^2$ and its creation-operator analogue. The corresponding sums without logarithms are finite by finite energy. These bounds imply the claimed absolute summability of K and vK .

In matrix coordinates, (5.5) is

$$q_j(Z) = \sum_{l,r} K_{lr} \overline{d'_{j,lr}} Z_{lr}.$$

By (5.2) and (5.3),

$$\frac{K_{lr}}{h\ell(p_l, p_r)} = b(v) - (r + \tfrac{1}{2})v = (1 - v)F(v). \quad (5.7)$$

Hence the representing vector in the F inner product is $(1 - v)d'_j$, and its squared norm is the sum in (5.6). That sum is finite by the preceding estimates. Finally, $F(0) = 1/h$, so orthogonality to I in the F inner product is exactly zero ρ -mean. The representing vector is orthogonal to I because $q_j(I) = 0$. Restricting to the centered space therefore does not change the dual norm. $\square$

Lemma 5.3 (Weak convolution identity). *Let $\mu_\gamma = \sum_i \sqrt{\lambda_i} \mu_i$ be the raw output mean. For every bounded output observable Z ,*

$$q_{0,j}(Z) = \sum_i w_i q_{i,j}(L_i Z) + \kappa(1 - \kappa) \overline{(\mu_\gamma)_j} \text{Tr}[(\gamma - \tau_{r_0})Z]. \quad (5.8)$$

Proof. The thermal defect is zero on τ_r , since $(r+1)d\tau_r = r\tau_r d$ by its geometric number weights. For the raw output, define $q_{\gamma,j}$ by (5.5) using its mean and reference parameter r_0 . The terms with coefficient 1 in the identity

$$q_{\gamma,j}(Z) = \sum_{s=1,2,D} \sqrt{\lambda_s} q_{s,j}(T_s Z)$$

agree by $d_{0,j} = \sum_s \sqrt{\lambda_s} d_{s,j}$. The remaining terms are expected commutators. After rotation, $d_{s,j}^\dagger$ is $\sqrt{\lambda_s} d_{0,j}^\dagger$ plus an operator on discarded modes. The weak commutator of that discarded operator with a retained observable has zero trace. Thus its retained coefficient is $\sum_s \lambda_s r_s = r_0$, as required.

These statements use only finite energy: all one-mode-factor products with the state are trace class as above. A discarded operator can first be cut off in its own number basis; its bounded cutoff commutes with $Z \otimes I$, so trace cyclicity applies. The cutoff products converge in trace norm by the same Hilbert–Schmidt factorization. This justifies the weak commutator and passage between joint, reduced, and sliced traces, without assigning an operator domain to $[d_{0,j}^\dagger, Z]$.

The D term vanishes. Since the mean of the replaced output is $(1 - \kappa)\mu_\gamma$, direct centering gives

$$q_{0,j} = (1 - \kappa)q_{\gamma,j} + \kappa(1 - \kappa) \overline{(\mu_\gamma)_j} \text{Tr}[(\gamma - \tau_{r_0}) \cdot].$$

The first term is $\sum_i w_i q_{i,j} \circ L_i$, since $q_{i,j}(I) = 0$. □

Proposition 5.4 (Defect estimate). *Set*

$$A_*^2 = \sum_i w_i \|q_i\|_{F_i,*}^2, \quad M_*^2 = \sum_i w_i |\mu_i|^2, \quad C_0 = (h_0 \inf_{v \in \mathbb{R}} F_0(v))^{-1/2}.$$

Then

$$\|q_0\|_{F_0,*} \leq A_* + \sqrt{2\kappa} C_0 M_*. \quad (5.9)$$

In particular, the previously reserved choice of κ may be made small enough that

$$\|q_0\|_{F_0,*}^2 \leq A_*^2 + \frac{\varepsilon}{2} (A_*^2 + M_*^2) \quad (5.10)$$

holds uniformly in ζ and the minimizing states.

Proof. By Proposition 5.1 and Cauchy–Schwarz, the dual norm of the first term in (5.8), summed over the modes, is at most A_* . For the second term put

$$J(Z) = \sqrt{\kappa(1 - \kappa)} \text{Tr}[(\gamma - \tau_{r_0})Z].$$

The variance between the two branches of the mixture gives $|J(Z)|^2 \leq \|Z\|_{R_{0,+},0}^2$ and $|J(Z)|^2 \leq \|Z\|_{R_{0,-},0}^2$ on centered tests. The two-weight logarithmic-mean operation proved in Section 3.1 applies to this scalar-valued map, with both target weights equal to 1. Since $\ell(R_{0,+}, R_{0,-}) = 1/h_0$, it yields

$$|J(Z)|^2 \leq \|Z\|_0^2 / h_0 \leq C_0^2 \|Z\|_{F_0}^2.$$

Moreover, $(1 - \kappa)|\mu_\gamma|^2 \leq 2 \sum_i w_i |\mu_i|^2 = 2M_*^2$. The triangle inequality now proves (5.9). Using Young's inequality with coefficient $\varepsilon/2$, it is enough to impose

$$0 < \kappa \leq \frac{\varepsilon^2}{4(\varepsilon + 2)C_0^2}$$

to obtain (5.10). The right-hand side depends only on the reference parameters and ε , which were fixed before κ ; this requirement is therefore compatible with the parameter choice in Proposition 4.2. $\square$

6 Stationarity and the thermal limit

For each label s , consider the thermal relaxation expression

$$\delta_s = \mathcal{L}_s \rho_s, \quad \mathcal{L}_s \rho = \sum_{j=1}^n [(r_s + 1)\mathcal{D}[d_{s,j}]\rho + r_s \mathcal{D}[d_{s,j}^\dagger]\rho], \quad \mathcal{D}[d]\rho = d\rho d^\dagger - \frac{1}{2}\{d^\dagger d, \rho\}. \quad (6.1)$$

This is the standard bosonic thermal-relaxation, or quantum Ornstein–Uhlenbeck, generator; its entropy production has been studied in [23, 5, 9]. Here we use it as a trace-class expression at the regularized minima. The proof requires the weighted trace identities established below, rather than a log-Sobolev inequality or differentiation of entropy along an unbounded semigroup. The geometric weights give $\mathcal{L}_s \tau_{r_s} = 0$.

6.1 Weighted traces and the exact stationarity identity

Lemma 6.1 (Generator covariance and logarithmic pairings). *At a minimizing pair, δ_s is traceless and $W_s^2 \delta_s W_s^2$ is trace class. If Φ_i is the raw channel with the other two input states fixed, extended linearly to trace-class operators, then*

$$\delta_0 = (1 - \kappa) \sum_i \Phi_i(\delta_i). \quad (6.2)$$

The traces against the forms H_s and W_s^4 are well defined, and

$$\text{Tr } \delta_0 H_0 = (1 - \kappa) \sum_i \text{Tr } \delta_i T_i H_0. \quad (6.3)$$

Proof. First consider finite total-number cutoffs, where all products in the calculation are finite rank. This covariance calculation is linear in the joint input and applies to arbitrary finite-number-supported joint operators, without a product assumption. The sum over the three ports of the coefficient-1 terms $\mathcal{D}[d_{s,j}]$ is invariant under the real orthogonal mode rotation. Terms acting only on discarded modes vanish under partial trace. The remaining heat expression is

$$\mathcal{D}[d]\rho + \mathcal{D}[d^\dagger]\rho = -\frac{1}{2}([d, [d^\dagger, \rho]] + [d^\dagger, [d, \rho]]).$$

Writing an original mode as its retained component plus discarded components, every commutator involving a discarded component disappears under partial trace. The retained part has coefficient λ_s . The resulting generator on the raw output therefore has parameter $\sum_s \lambda_s r_s = r_0$. We next justify this covariance calculation for the full input state.

Here is a weighted trace argument extending the finite-cutoff calculation. Suppress a label and put $T = W^3 \rho W^3$, a positive trace-class operator by the finite seventh moment. Each term of $W^2 \mathcal{L} \rho W^2$ is a finite sum of bounded-factor products around T . For example,

$$W^2 d \rho d^\dagger W^2 = (W^2 d W^{-3}) T (W^{-3} d^\dagger W^2), \quad W^2 d^\dagger d \rho W^2 = (d^\dagger d W^{-1}) T W^{-1}.$$

The outer factors are bounded because a mode shifts number by one and has coefficients growing as the square root of number. The creation terms satisfy the same estimates. For the number cutoff P_K , $W^3 P_K \rho P_K W^3 = P_K T P_K \rightarrow T$ in trace norm. Hence the cutoff generator expressions converge in the W^2 -sandwiched trace norm. In particular they have trace-zero limits.

The same estimates hold on the joint input using $W_{\text{tot}} = I + N_1 + N_2 + N_D$. The joint input has finite seventh moment, and the rotation preserves total-number sectors. Partial trace is continuous in the required weighted trace norms: for $k = 3$ on cutoff states and $k = 2$ on generator expressions, insert the bounded factor $(W_0^k \otimes I) W_{\text{tot}}^{-k}$, of norm at most one, on both sides before taking the partial trace. On the full product input the fixed thermal port has zero generator, as does the replacement thermal. Consequently the covariance just proved reduces to (6.2).

The form bound $0 \leq H_s \leq C_s W_s^4$ makes $W_s^{-2} H_s W_s^{-2}$ bounded. Its pairing with $W_s^2 \delta_s W_s^2$ defines the logarithmic trace; W_s^4 is treated the same way. For the slicing assertion, decompose the self-adjoint trace-class operator $W_i^2 \delta_i W_i^2$ into its positive and negative parts and conjugate back by W_i^{-2} . This expresses δ_i as a difference of positive operators with finite fourth moment. The positive slicing identity for $T_i H_0$ applies to each part, proving (6.3). $\square$

Define $\sigma_s = \text{Tr } \delta_s H_s / h_s$. Pair the exact Hamiltonian identity (4.4) with δ_i and sum over i . The scalar terms vanish by tracelessness, and Lemma 6.1 gives

$$0 = \sigma_0 - (1 + \varepsilon) \sum_i w_i \sigma_i + \varepsilon \sum_i w_i \text{Tr } \delta_i N_i + \zeta \sum_i \text{Tr } \delta_i W_i^4. \quad (6.4)$$

This is an algebraic consequence of the Gibbs identity. It requires neither that $\rho_i + t \delta_i$ remain positive for two-sided t , nor that entropy be differentiated along this unbounded generator.

Lemma 6.2 (Entropy production identity). *The quantities in (6.4) satisfy*

$$\sigma_s = \|q_s\|_{F_{s,*}}^2 - (e'_s - n r_s). \quad (6.5)$$

Proof. In an eigenbasis of ρ_s , direct evaluation of the two dissipators gives

$$\sigma_s = - \sum_{j,l,r} v_s(l,r) ((r_s + 1) p_r - r_s p_l) |(d'_{s,j})_{lr}|^2. \quad (6.6)$$

For example, the contribution of $\mathcal{D}[d] \rho$ to the trace against H is $\sum_{l,r} p_r (-\log p_l + \log p_r) |d_{lr}|^2$. The creation term is obtained by interchanging l, r ; centering d changes only diagonal entries, where $v_s(l, l) = 0$.

All these expansions are legitimate. Lemma 5.2 controls the crossed and uncrossed logarithmic sums absolutely. In the anticommutator terms, $d^\dagger d \psi_r$ and $d d^\dagger \psi_r$ belong to $D(W^2)$ because $\psi_r \in D(W^{7/2})$. The form pairing with $H \psi_r = (-\log p_r) \psi_r$ is therefore valid. For the eigenbasis truncations $\rho^{(m)} = \sum_{r \leq m} p_r |\psi_r\rangle \langle \psi_r|$, we have

$$W^3 \rho^{(m)} W^3 = \sum_{r \leq m} p_r |W^3 \psi_r\rangle \langle W^3 \psi_r| \longrightarrow W^3 \rho W^3 \quad \text{in trace norm.}$$

The bounded-factor expansions in Lemma 6.1 therefore justify the eigenvector expansions in the required weighted trace norm as well.

Subtracting (6.6) from (5.6) leaves

$$\sum_{j,l,r} ((r_s + 1)p_r - r_s p_l) |(d'_{s,j})_{lr}|^2 = (r_s + 1)e'_s - r_s(e'_s + n) = e'_s - nr_s,$$

where the canonical commutation relations give the additional n . This proves (6.5). $\square$

6.2 Cancellation and convergence to the thermal inputs

The number shifts in (6.1) give

$$\text{Tr } \delta_i N_i = nr_i - \text{Tr } \rho_i N_i = -(e'_i - nr_i) - |\mu_i|^2. \quad (6.7)$$

They also give

$$\begin{aligned} \text{Tr } \delta_i W_i^4 &= \text{Tr } \rho_i P_i(N_i), \\ P_i(N_i) &= (r_i + 1)N_i((W_i - 1)^4 - W_i^4) + r_i(N_i + n)((W_i + 1)^4 - W_i^4). \end{aligned} \quad (6.8)$$

This polynomial has leading term $-4N_i^4$; hence its supremum on $N_i \in \{0, 1, \dots\}$ is finite. In particular,

$$P := \sum_i \text{Tr } \delta_i W_i^4 \leq C, \quad (6.9)$$

with C independent of ζ and the minimizing states.

Independence across the input ports and $\mu_D = 0$ imply

$$e'_\gamma = \sum_i \lambda_i e'_i + \lambda_D nr_D, \quad \mu_\gamma = \sum_i \sqrt{\lambda_i} \mu_i.$$

Taking account of the two branches of the replacement gives the exact centered-energy identity

$$e'_0 - nr_0 = \sum_i w_i (e'_i - nr_i) + \kappa(1 - \kappa) |\mu_\gamma|^2. \quad (6.10)$$

Insert (6.5), (6.7), and (6.10) into (6.4). All centered-energy terms cancel, yielding, before discarding any term,

$$0 = \|q_0\|_{F_0,*}^2 - (1 + \varepsilon)A_*^2 - \varepsilon M_*^2 - \kappa(1 - \kappa) |\mu_\gamma|^2 + \zeta P. \quad (6.11)$$

Now Proposition 5.4 and (6.9) imply

$$0 \leq -\frac{\varepsilon}{2}(A_*^2 + M_*^2) + C\zeta. \quad (6.12)$$

Thus, as $\zeta \downarrow 0$, both input means tend to zero and both lists of input defect functionals tend to zero in their dual norms.

The uniform energy bound in Proposition 4.2 and Lemma 2.2 provide a subsequence along which both inputs converge in trace norm, say to $\hat{\rho}_i$. To identify these limits, fix a matrix Z supported on a finite block of the number basis. By Proposition 5.1 and the arithmetic bound for the logarithmic mean, the centered version of Z has F_i norm at most a fixed constant times $\|Z\|$, uniformly in the minimizing state. Hence $q_{i,j}(Z) \rightarrow 0$ by (6.12). Replacing $d'_{i,j}$ by $d_{i,j}$ changes this functional by $\overline{(\mu_i)_j} \text{Tr } \rho_i Z$, which also tends to zero. Both $d'_{i,j}^\dagger Z$ and $Z d'_{i,j}^\dagger$ are bounded finite-rank operators for this fixed test. Trace-norm convergence therefore passes the uncentered trace identity to the limit, giving

$$(r_i + 1)d_{i,j}\hat{\rho}_i = r_i\hat{\rho}_i d_{i,j} \quad \text{as number-basis matrices, for every } j. \quad (6.13)$$

No convergence of energy or of unbounded first moments is used here.

For completeness, (6.13) determines the whole n -mode state. If multi-indices $\mathbf{p}, \mathbf{b}$ satisfy $p_j > 0$, its matrix entries obey

$$(r_i + 1)\sqrt{p_j}(\hat{\rho}_i)_{\mathbf{p}, \mathbf{b}} = r_i\sqrt{b_j}(\hat{\rho}_i)_{\mathbf{p}-\mathbf{e}_j, \mathbf{b}-\mathbf{e}_j},$$

with zero right-hand side when $b_j = 0$. If $p_j > b_j$, iteration kills the entry; if $p_j < b_j$, self-adjointness does the same. Thus all off-diagonal entries vanish. Diagonal entries have ratio $r_i/(r_i + 1)$ whenever any one coordinate is increased by one. Trace one fixes the normalization, so $\hat{\rho}_i = \tau_{r_i}$. This conclusion excludes internal correlations as well as off-diagonal coherences in the limit; no product assumption within an input was made.

The raw output now converges to τ_{r_0} by continuity of the channel and Proposition 2.3; the replaced output has the same limit. Uniformly bounded energies give convergence of all three entropies by Lemma 2.2. The entropy portion of J_ζ in (4.1) consequently tends to zero. Its relative-entropy and fourth-moment penalties are nonnegative, so

$$\liminf_{\zeta \downarrow 0} J_\zeta \geq 0$$

along the selected subsequence. This contradicts the uniform strict negative upper bound for the minima. It is unnecessary to prove that either penalty tends to zero, or to send any of the other auxiliary parameters to zero.

The strict violation assumed in Section 4 is therefore impossible. Since the reduction there preserved every arbitrary finite-energy input pair at each fixed finite n , this proves Theorem 1.1 for $0 < \eta < 1$. At $\eta = 0$ or 1 the retained state is the corresponding input and the asserted inequality is equality.

7 A degraded pure-loss broadcast capacity region

Write $\mathcal{L}_\xi = \mathcal{E}_{\xi,0}$ for the pure-loss attenuator. Consider the memoryless broadcast channel with one input mode per use, passively split with vacuum auxiliary inputs between receivers B, C and an inaccessible loss output. Let η_B, η_C be the power transmissivities to the two receivers, so the marginal channels are $\mathcal{L}_{\eta_B}$ and $\mathcal{L}_{\eta_C}$. We assume

$$0 \leq \eta_C < \eta_B, \quad \eta_B + \eta_C \leq 1.$$

For an n -use code, the messages $m_B \in \{1, \dots, M_B\}$ and $m_C \in \{1, \dots, M_C\}$ are independent and uniform. The encoder may assign to each pair an arbitrary finite-energy n -mode state $\rho_{m_B, m_C}^{(n)}$, including states entangled across uses, subject to

$$\frac{1}{M_B M_C} \sum_{m_B, m_C} \text{Tr} \left(\rho_{m_B, m_C}^{(n)} \sum_{j=1}^n a_j^\dagger a_j \right) \leq n\bar{N}. \quad (7.1)$$

Thus the photon constraint is averaged over uses and the entire codebook, not imposed separately on each codeword. Each receiver uses an arbitrary collective measurement on its own n output modes to recover its intended message, with vanishing average error. The receivers do not cooperate, and there is no feedback.

Corollary 7.1 (Degraded two-receiver pure-loss broadcast capacity). *Let $0 \leq \bar{N} < \infty$, $0 \leq \eta_C < \eta_B$, and $\eta_B + \eta_C \leq 1$. The classical capacity region for the channel and coding model above, with rates*

in nats per use, is the closed convex hull of the nonnegative pairs (R_B, R_C) satisfying, for some $0 \leq \beta \leq 1$,

$$\begin{aligned} R_B &\leq g(\eta_B \beta \bar{N}), \\ R_C &\leq g(\eta_C \bar{N}) - g(\eta_C \beta \bar{N}), \end{aligned} \quad (7.2)$$

where g is the natural-log function in (1.1).

Proof. Converse. Set $\lambda = \eta_C/\eta_B \in [0, 1]$. Composition of pure-loss attenuators gives $\mathcal{L}_{\eta_C} = \mathcal{L}_\lambda \circ \mathcal{L}_{\eta_B}$. This is a degrading channel for the receiver marginals, implemented with a new independent vacuum port; it does not assert independence of the physical B and C outputs.

For every finite n and every finite-energy n -mode state σ , Theorem 1.1 with an independent vacuum port, equivalently Corollary 1.2 at $N_B = 0$, gives

$$S(\mathcal{L}_\lambda^{\otimes n}(\sigma)) \geq ng \left(\lambda g^{-1} \left(\frac{S(\sigma)}{n} \right) \right). \quad (7.3)$$

This supplies the finite-block vacuum-port inequality required from Strong Conjecture 2 in Guha, Shapiro, and Erkmen's broadcast analysis [21, Appendix A] on all states needed here. It does not require σ to be a product across modes.

We follow the converse of Guha, Shapiro, and Erkmen [21], giving the averaging and information argument for the present coding model. For a fixed n -use code, let σ_{mk}^Y be the output at receiver $Y \in \{B, C\}$ for messages $m = m_B$, $k = m_C$, and set

$$\sigma_k^Y = \frac{1}{M_B} \sum_m \sigma_{mk}^Y, \quad \bar{\sigma}^C = \frac{1}{M_C} \sum_k \sigma_k^C, \quad \bar{N}_k = \frac{1}{nM_B} \sum_m \text{Tr} \rho_{m,k}^{(n)} N.$$

The codebook constraint gives $M_C^{-1} \sum_k \bar{N}_k \leq \bar{N}$. For every finite-energy input, a vacuum attenuator satisfies

$$\text{Tr}(\mathcal{L}_\xi^{\otimes n}(\rho)N) = \xi \text{Tr} \rho N :$$

expanding each output number operator, the independent vacuum has zero number and first moments, so its number term and both cross terms vanish. This calculation allows arbitrary entanglement across input modes. Thus σ_k^B has energy $n\eta_B \bar{N}_k$, while $\bar{\sigma}^C$ has energy at most $n\eta_C \bar{N}$. All conditional states have finite energy and entropy, and linearity and degradation give $\sigma_k^C = \mathcal{L}_\lambda^{\otimes n}(\sigma_k^B)$.

Let $p_{B,n}, p_{C,n}$ be the intended-message average error probabilities, and write $R_{Y,n} = n^{-1} \log M_Y$. Fano's inequality and the Holevo bounds used in the cited converse give

$$\begin{aligned} (1 - p_{B,n})R_{B,n} &\leq \frac{1}{nM_C} \sum_k S(\sigma_k^B) + \frac{H_2(p_{B,n})}{n}, \\ (1 - p_{C,n})R_{C,n} &\leq \frac{S(\bar{\sigma}^C)}{n} - \frac{1}{nM_C} \sum_k S(\sigma_k^C) + \frac{H_2(p_{C,n})}{n}, \end{aligned}$$

where H_2 is the binary entropy. For the first bound, reveal the independent weak message as side information to the strong decoder and discard the nonnegative entropies of the individual output states in the conditional Holevo quantity. This only enlarges the information used for the upper bound; it does not change the decoder required by the model. The second bound is the Holevo bound for the weak message.

Assume $\bar{N} > 0$ for now, and put

$$u_{k,n} = \frac{S(\sigma_k^B)}{n}, \quad \bar{u}_n = \frac{1}{M_C} \sum_k u_{k,n}.$$

The energy–entropy bound and concavity of g , whose second derivative is $-1/[r(1+r)]$ for $r > 0$, imply

$$0 \leq \bar{u}_n \leq \frac{1}{M_C} \sum_k g(\eta_B \bar{N}_k) \leq g(\eta_B \bar{N}).$$

There is therefore a code-dependent $\beta_n \in [0, 1]$ with $\bar{u}_n = g(\eta_B \beta_n \bar{N})$. Set $\phi_\lambda(u) = g(\lambda g^{-1}(u))$. This function is convex. Indeed, for $0 < \lambda < 1$ and $r > 0$, differentiation using g and $h = g'$ gives

$$\phi''_\lambda(g(r)) = \frac{\lambda[(1+\lambda r)h(\lambda r) - (1+r)h(r)]}{r(1+r)(1+\lambda r)h(r)^3} \geq 0,$$

because $t \mapsto (1+t)h(t)$ has derivative $h(t) - 1/t < 0$. Continuity includes $u = 0$, and $\phi_0 = 0$. Applying (7.3) to each conditional state and then Jensen's inequality gives

$$\frac{1}{nM_C} \sum_k S(\sigma_k^C) \geq \frac{1}{M_C} \sum_k \phi_\lambda(u_{k,n}) \geq \phi_\lambda(\bar{u}_n) = g(\eta_C \beta_n \bar{N}).$$

Also $S(\bar{\sigma}^C) \leq ng(\eta_C \bar{N})$. Substituting these bounds into the information inequalities yields

$$\begin{aligned} (1 - p_{B,n})R_{B,n} &\leq g(\eta_B \beta_n \bar{N}) + \frac{H_2(p_{B,n})}{n}, \\ (1 - p_{C,n})R_{C,n} &\leq g(\eta_C \bar{N}) - g(\eta_C \beta_n \bar{N}) + \frac{H_2(p_{C,n})}{n}. \end{aligned}$$

For any achieved rate pair, take a subsequence on which β_n converges in $[0, 1]$. Continuity of g , the rate lower limits, and vanishing decoding errors give the outer bound in (7.2) with the limiting β .

Achievability. We use the coherent Gaussian superposition ensemble of Guha, Shapiro, and Erkmén [21, Section IV]. To justify its bosonic limit and the photon constraint, we start with the finite-alphabet superposition construction in the proof of Yard, Hayden, and Devetak's Theorem 1 [33, Section II.A], expressed here in nats. For a finite classical input alphabet, finite-dimensional quantum outputs, and a finite law $p(t, x)$, that construction sends a common message to both receivers and a private message to B at the strict rates

$$R_B < \sum_t p(t) \chi(p(x | t), \omega_x^B), \quad R_C < \min_{Y=B,C} \chi(p(t), \omega_t^Y),$$

where $\omega_t^Y = \sum_x p(x | t) \omega_x^Y$ and

$$\chi(p_j, \omega_j) = S\left(\sum_j p_j \omega_j\right) - \sum_j p_j S(\omega_j).$$

We use the common message as the message for C ; receiver B may discard its decoded common label. Degradation and data processing make the minimum in the common-message bound equal to the C quantity.

Assume first that $\bar{N} > 0$ and $\eta_C > 0$. Fix $0 < N' < \bar{N}$ and $0 < \beta < 1$. Let T, Z be independent centered circular complex Gaussian variables of variances $(1-\beta)N'$ and $\beta N'$, respectively. Partition the disk $\{|z| \leq \ell\}$ into finitely many sets of diameter at most $1/\ell$, with its complement as one further cell, and let T_ℓ, Z_ℓ be the corresponding conditional expectations of T, Z . For either $X = T, Z$ and its corresponding X_ℓ ,

$$\mathbb{E}|X - X_\ell|^2 \leq \ell^{-2} + \mathbb{E}[|X|^2 \mathbf{1}_{\{|X| > \ell\}}] \rightarrow 0.$$

They remain centered and independent, and conditional expectation contracts second moments. Hence

$$\mathbb{E}|T_\ell + Z_\ell|^2 = \mathbb{E}|T_\ell|^2 + \mathbb{E}|Z_\ell|^2 \leq N'.$$

For a symbol $x = (t, z)$ of their finite product alphabet, transmit the coherent state $|t + z\rangle$. The finite law in the coding construction is $p(t, (t', z)) = \mathbf{1}_{\{t=t'\}} p_{T_\ell}(t) p_{Z_\ell}(z)$. At receiver transmissivity $\xi \in \{\eta_B, \eta_C\}$, put

$$\theta_{\ell, \xi} = \mathbb{E}|\sqrt{\xi}Z_\ell\rangle\langle\sqrt{\xi}Z_\ell|, \quad \omega_{\ell, \xi} = \mathbb{E}|\sqrt{\xi}(T_\ell + Z_\ell)\rangle\langle\sqrt{\xi}(T_\ell + Z_\ell)|.$$

Conditioned on $T_\ell = t$, the receiver state is a unitary displacement of $\theta_{\ell, \xi}$, and the output for each symbol is pure. The private-message Holevo quantity at B and the common-message Holevo quantity at C are therefore

$$S(\theta_{\ell, \eta_B}), \quad S(\omega_{\ell, \eta_C}) - S(\theta_{\ell, \eta_C}). \quad (7.4)$$

The coherent-projector identity

$$\| |u\rangle\langle u| - |v\rangle\langle v| \|_1 = 2\sqrt{1 - e^{-|u-v|^2}} \leq 2|u-v|$$

and Cauchy-Schwarz show that, in trace norm, $\theta_{\ell, \xi}$ tends to the one-mode thermal state of mean photon number $\xi\beta N'$, and $\omega_{\ell, \xi}$ to the one-mode thermal state of mean photon number $\xi N'$. Their mean photon numbers are bounded by $\xi\beta N'$ and $\xi N'$, respectively. Lemma 2.2 thus gives convergence of the quantities in (7.4) to

$$g(\eta_B\beta N'), \quad g(\eta_C N') - g(\eta_C\beta N').$$

For each fixed ℓ , the finitely many coherent receiver vectors span finite-dimensional spaces. The physical channel on these symbols is therefore exactly a finite classical-input channel with such output spaces. Decoding measurements on those spaces extend to Fock space without changing their action on the transmitted states.

It remains to enforce (7.1) on a deterministic code. For the fixed finite alphabet let

$$c(x) = |t + z|^2, \quad c_t = \sum_x p(x | t) c(x), \quad \bar{c} = \sum_t p(t) c_t \leq N', \quad c_{\max} = \max_x c(x).$$

Fix a target rate pair strictly below the two finite Holevo bounds. Choose the type tolerance $\delta > 0$ small enough for this rate backoff and so that $\bar{c} + \delta c_{\max} \leq N'' < \bar{N}$ for some N'' . The cited superposition construction supplies a sequence of fixed outer codes whose words have a common type P_n at block length n , with $\|P_n - p_T\|_1 \leq \delta$, where $p_T(t) = p(t)$. The random inner letters in the corresponding blocks have law $p(x | t)$, and the complete words are obtained by permutations. With each outer code fixed, all expectations and probabilities below are over these inner codebooks. Thus the photon cost C_n per use, averaged over the entire random codebook, satisfies

$$\mathbb{E}C_n = \sum_t P_n(t) c_t \leq \bar{c} + \delta c_{\max} \leq N''.$$

Before its final message expurgation, the cited construction also gives a vanishing expected bound on its average joint decoding error over the same inner codebooks. Let E_n be the sum of the two intended-message average errors, with B 's decoded common label discarded. Each is bounded by that joint error, so, after absorbing a factor of two, $\mathbb{E}E_n \leq e_n$ for a positive sequence $e_n \rightarrow 0$. We

use this stage because its average-error estimate suffices for the stated criterion and its full product message set is the one over which $\mathbf{C}_n$ was computed. Markov's inequality gives

$$\Pr\{\mathbf{C}_n > \bar{N}\} + \Pr\{\mathbf{E}_n > \sqrt{e_n}\} \leq \frac{N''}{\bar{N}} + \sqrt{e_n} < 1$$

for all sufficiently large n . Hence a deterministic code satisfies the entire-codebook photon constraint and has vanishing average error. It retains the full uniform independent product message set. Each codeword is a product of finitely many coherent symbols and has finite energy, and $\mathbf{E}_n$ controls each receiver's intended-message error.

For $0 < \beta < 1$, fix a rate pair strictly below the corresponding Gaussian pair in (7.2). First choose $N' < \bar{N}$ close enough that its Gaussian pair still exceeds the target, and then choose a finite ℓ whose Holevo pair does so. For this fixed alphabet choose the type tolerance and cost slack as above, and only then let the block length grow. Taking closure reaches the boundary and includes $\beta = 0, 1$, and time sharing among codes satisfying the same photon bound gives the closed convex hull.

If $\bar{N} = 0$, every codeword is the vacuum and both rates are zero. If $\eta_C = 0$, receiver C always receives the vacuum and the region reduces to $R_C = 0$, $R_B \leq g(\eta_B \bar{N})$, the single-receiver pure-loss capacity [18]. Both endpoints are included in (7.2). $\square$

Corollary 7.1 concerns only the stated two-receiver pure-loss classical coding model. It gives no noisy-broadcast, amplifier, wiretap, quantum-capacity, feedback, or receiver-cooperation statement.

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